

Response to Reviewers

Windows Integration into the Green Metrics Tool: Energy Measurement of Native Windows Applications via RAPL

We thank all three reviewers for their careful and constructive reports. The reviews converged on one theme — the breadth of the evaluation and the basis on which the reported values can be judged — and the revision is built around it.

The main change: the native Windows implementation of GMT, which the reviewed version named only as an outlook in §5.4, has been implemented and evaluated. The reviewed version also reported a single measurement run, and stated in §5.3 that repeated runs under identical conditions would be required to establish reproducibility. Those runs have now been carried out. The revised paper therefore presents two hosting designs, each with a ten-run reproducibility study, and compares them directly.

What is new since the reviewed version

- **Second Design 2 (native Windows).** The GMT runner now executes as a native Windows process with Docker Desktop handling container orchestration, removing the runner's dependency on WSL2.
- **Ten-run reproducibility studies (n = 10) for both designs,** extending the single run. Total-energy coefficients of variation are 1.6 % (WSL2-bridged) and 2.2 % (native Windows) for `cpu_package`, and at most 2.7 % across all measured domains.
- **A direct comparison of the two hosting designs,** including per-phase energy and aligned power traces.
- **Presentation.** All figures are now labelled in English, the experimental configuration table has been expanded, and the paper has been brought to 10 pages for the LNI short-paper limit.

Reviewer 1

Plausibility of the results; comparison against a Linux machine or other comparative measurements. This was the reviewed version's weakest point, and we have addressed it in two ways. First, plausibility is no longer argued qualitatively: each design was measured ten times, and we report means, standard deviations and coefficients of variation per RAPL domain, so run-to-run stability can be judged directly. Second, we added a comparative measurement in the sense you asked for — a second, independently implemented acquisition path (Windows EMI) running alongside Scaphandre/MSR, which agrees to within 0.1 %. We have also been explicit about what that does *not* show: because both paths ultimately surface the same RAPL package counter, this is a consistency check on the software pipeline rather than an external validation. The limitations section now states plainly that no external ground-truth validation was performed, and names validation against a physical reference power meter as future work. A comparison against a similarly configured Linux machine has not been carried out within this revision.

EMIS as the measured object. We agree, and we have moderated the claims in this direction rather than expanding them. Measuring an EMIS itself is not primarily an instrumentation problem but a scenario-design problem: an EMIS is interactive and functionally heterogeneous, so a deterministic batch render of the kind used here does not represent it. Defining reproducible standard usage scenarios for such systems is a separate piece of work, and we did not want to imply otherwise on the basis of the evidence presented.

Abbreviations. RAPL and ACPI are now expanded on first use. The passage containing eBPF was removed during the shortening, so that abbreviation no longer occurs.

Fig. 1 could be larger. Figure sizes are constrained by the 10-page limit, so Fig. 1 has not been enlarged. The energy-per-phase figure has been rebuilt with substantially larger type, and its legend moved outside the plot area, where it previously occluded a data label.

Figure labelling not consistently in English. Corrected — all figure titles, axis labels and legends are now in English.

Reviewer 2

The contribution is small. The contribution has grown since the reviewed version: it now covers two hosting architectures rather than one, each evaluated with ten repeated runs, plus a second acquisition path and a comparison between the two designs. We have at the same time kept its boundaries explicit — the kernel driver and acquisition loop adopted from Scaphandre are identified as such throughout, and the novelty claim is stated narrowly: that no prior work integrates Windows RAPL measurement natively into a Docker-orchestrated energy benchmarking framework.

Is file-based coupling the right mechanism, or would a more efficient IPC method be better? This is a fair question and the rationale was not stated in the reviewed version. File-based hand-off is not a free design choice here: it is the contract that every GMT metric provider implements, and adopting it is what allows the integration to work without any modification to GMT's core components — one of the stated design goals — and allows the provider to be maintained upstream. It also carries no efficiency penalty in this setting: the acquisition binary emits one sample per 99 ms, and the output file is parsed only after the run has finished, so the coupling adds no runtime overhead to the measured workload. A lower-latency IPC channel would add a GMT-side dependency without improving either the sampling rate or measurement fidelity.

Figures in German. Corrected; see Reviewer 1 above.

No details about the evaluation platform, so the reported values cannot be judged. Agreed — the reviewed version gave little beyond the CPU model. The configuration table now specifies the renderer, scene, samples per pixel, resolution, render duration per design, operating system and container/WSL2 environment, CPU, RAM, RAPL sampling rate and measured domains. Beyond the platform description, the move from one run to ten with reported standard deviations and coefficients of variation is what we think most helps a reader judge whether the reported energies are plausible.

Reviewer 3

Validation limited to a single workload on one hardware platform; comparison against a reference implementation. Partially addressed. The evaluation now covers two configurations rather than one, with $n = 10$ each, and adds the second acquisition path described above. It remains a single hardware platform and a single, sustained CPU-bound workload class; this is stated explicitly in the limitations, and validation on further CPU architectures, non-CPU-bound workload classes and against an external power meter are named as future work rather than implied.

Some conclusions extend beyond the presented evidence; broader statements should be moderated or omitted. We agree and have moderated them. The comparison between the two hosting designs shows a 9.1 % difference in render energy and a 10.9 % difference in render duration. Rather than attributing this to a mechanism, the paper now reports it as observed in the tested configuration and states explicitly that it is not separable from a time-of-day confound, since the two sets of runs were collected at different times of day. The EMI agreement is no longer described as validating the measurement pipeline end to end. The suggestion that WSL2 isolation benefits measurement repeatability is hedged, since the WSL2 runs were also the quieter sessions. The broader statements about heterogeneous enterprise environments and ecological assessment have been reduced to what the evidence supports.

Related work could position the contribution more comprehensively. Background and related work have been merged into one section and condensed to meet the page limit, so this section is shorter rather than broader. The comparators that anchor the positioning — PowerAPI, Scaphandre, EnergiBridge and GMT — are retained, each with the specific dimension on which it differs from this work (platform coverage, instrumentation level, and whether it connects to a certification methodology).

Long sentences in §2.1 and §5.1. The §2.1 passage has been removed as part of the de-duplication. The §5.1 sentence has been split into several shorter statements ("From an EMIS perspective, two findings are relevant. First... Second...").

Repeated phrasing. "Hardware-level energy measurement" no longer occurs; "metric-provider architecture" now appears once.

Note on the benchmark configuration. For the repetition study the benchmark was re-run at 1440×1080 rather than 1920×1080 , which is why the absolute energy figures differ from those in the reviewed version. All values reported in the revision come from these new runs and have been cross-checked between text, tables and figures.